sense as the circular cross section of a sphere resides in a two-dimensional brane in Figure 22.2, and a spherical cross section of a hypersphere resides in a three-dimensional brane in Figure 22.3. So everything in Murph's bedroom, including Murph herself, is also inside the tesseract's back face.

When a light ray traveling out from Murph reaches the common edge of Murph's bedroom and the tesseract, it has two places to go: The ray can stay in our brane, traveling along route 1 of Figure 29.5 out an open door or into a wall where it is absorbed. Or the ray can stay in the tesseract, traveling along route 2 into and through the next tesseract face, and then onward to Cooper's eyes. Some of the ray's photons go along route 1; others go along route 2, bringing Cooper an image of Murph.

Now look at Figure 29.6, in which I restore the suppressed dimension. When Cooper looks through the right wall of his chamber, he sees into Murph's bedroom through its right wall (right white light ray). Looking through the left wall of his chamber, Cooper sees into Murph's bedroom through its left wall (left white light ray). Looking through his back wall, he sees into the bedroom through its back wall. Looking through his front wall (orange light ray), he sees into the bedroom through its front wall (though this is not obvious in Figure 29.6; can you explain why it is true?). Looking along the yellow ray, he sees down through her ceiling. Looking along the red ray, he sees up through her floor. To Cooper, as he changes his gaze from one direction to another to another, it seems like he is orbiting Murph's bedroom. (This is how Chris described it when he first showed me his complexified tesseract.)